

Portfolio Theory

Investment Policy

Nature of scheme	Information
Fund Manager	
Benchmark	BSE 200
Scheme Objective	Growth scheme (long-term growth)
Asset Allocation	80-100%: Equities & related securities. 0-20%: Debt & Money Market instruments
Investor Risk Profile	Medium to High
Investment Style	Growth
Investment Plans	Dividend Reinvestment & Growth
Variable Cost of Fund Management	2.25%
Fixed Cost of Fund Management	0.50%
Portfolio Rebalancing Calendar	Monthly

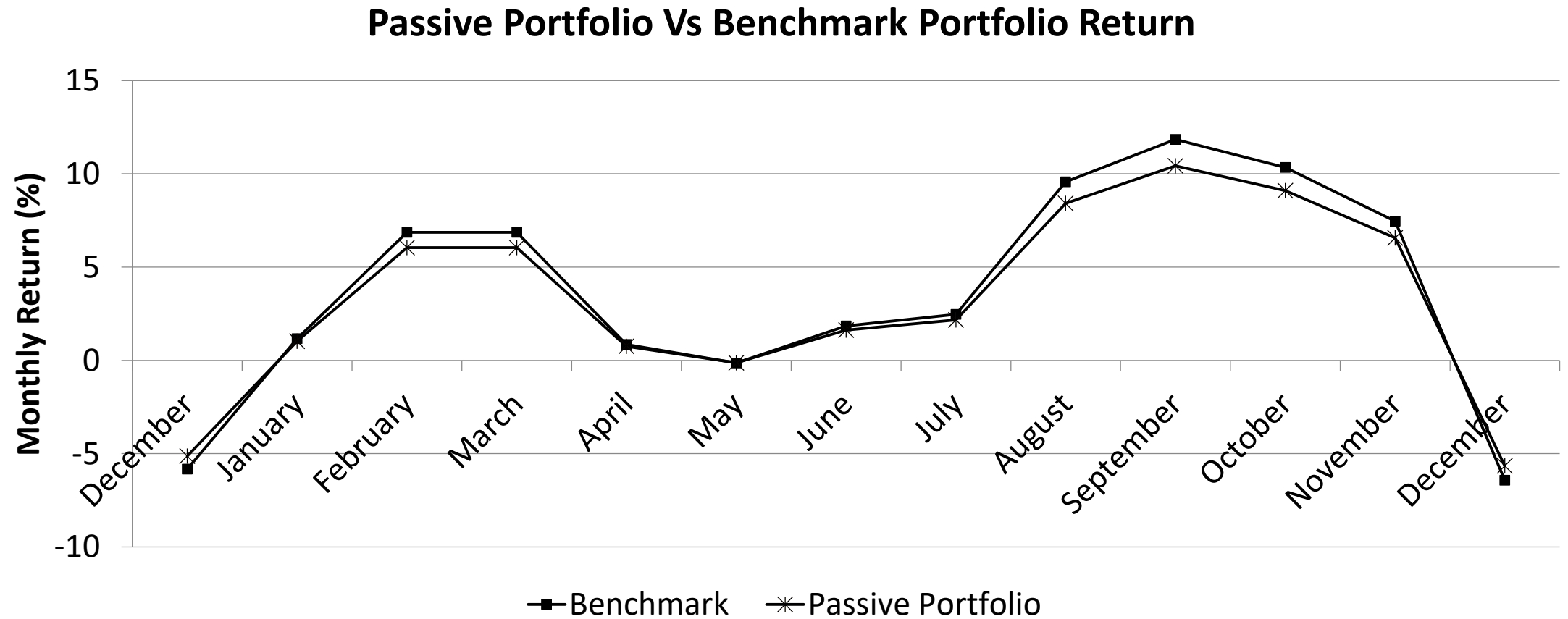
Benchmark Portfolios

- Benchmark portfolio represents the performance evaluation standard for portfolios before they are formed
- Benchmark portfolio are passive index portfolios
 - BSE Sensex, BSE 100, S & P Nifty
- Index portfolios represent a financial market performance
- Indices are mathematical formulations that are constructed based on a group of representative instruments to measure value movements in markets such as shares, bonds and commodities

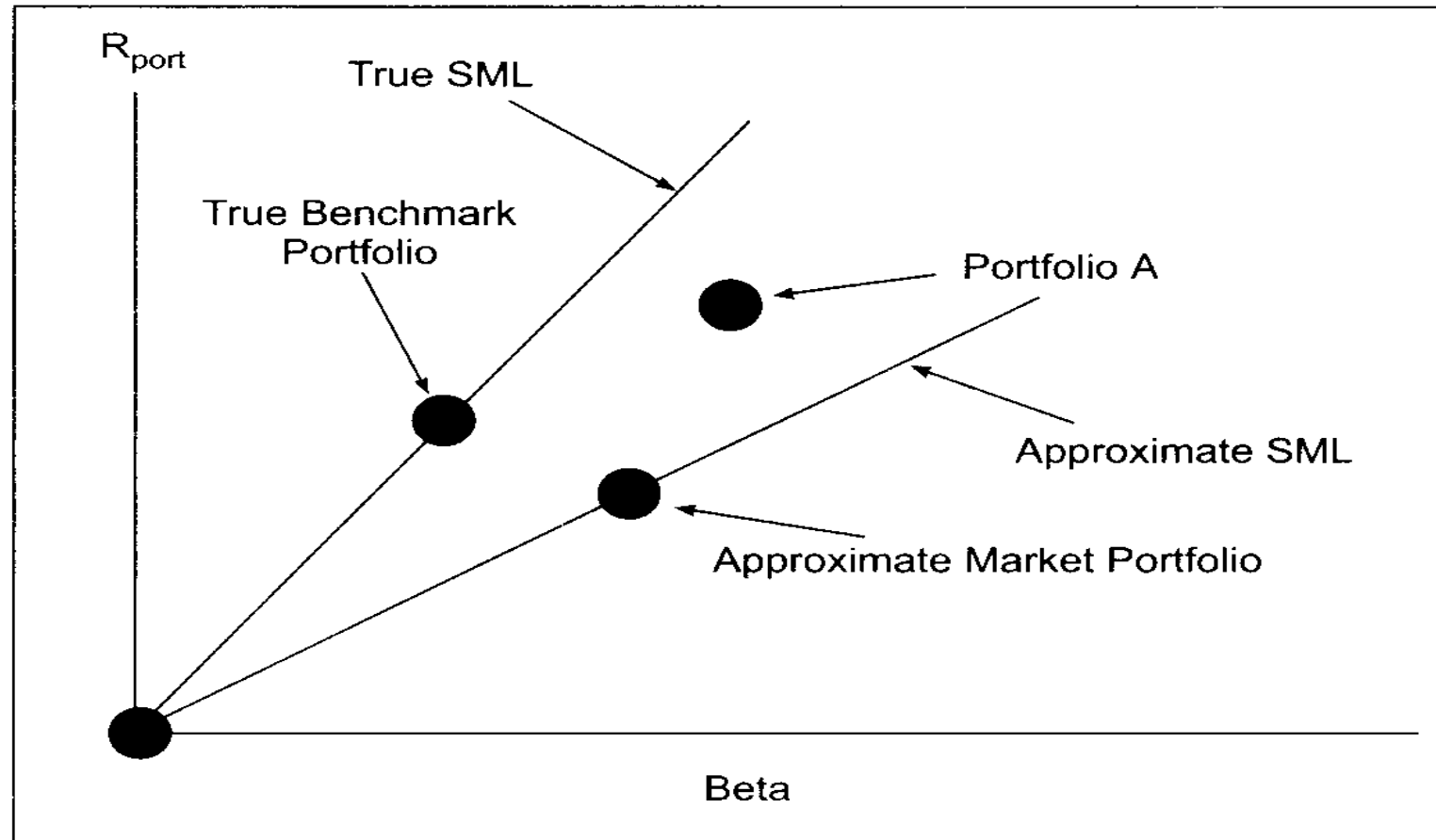
Characteristics of Benchmark / Passive Portfolios

- **Benchmark Portfolios:**
 - **Unambiguous**
 - **Investable**
 - **Measurable**
 - **Appropriate**
 - **Pre-specified**
- Passive portfolios do not require market monitoring. Buy and hold strategy outperforms a trade opportunity.

Passive Vs. Benchmark Portfolio Returns



Benchmark Errors



Investment Style

Investment Style			
Value	Blend / Cyclical	Growth	Size
			Large Market Cap
			Medium Market Cap
			Small Market Cap

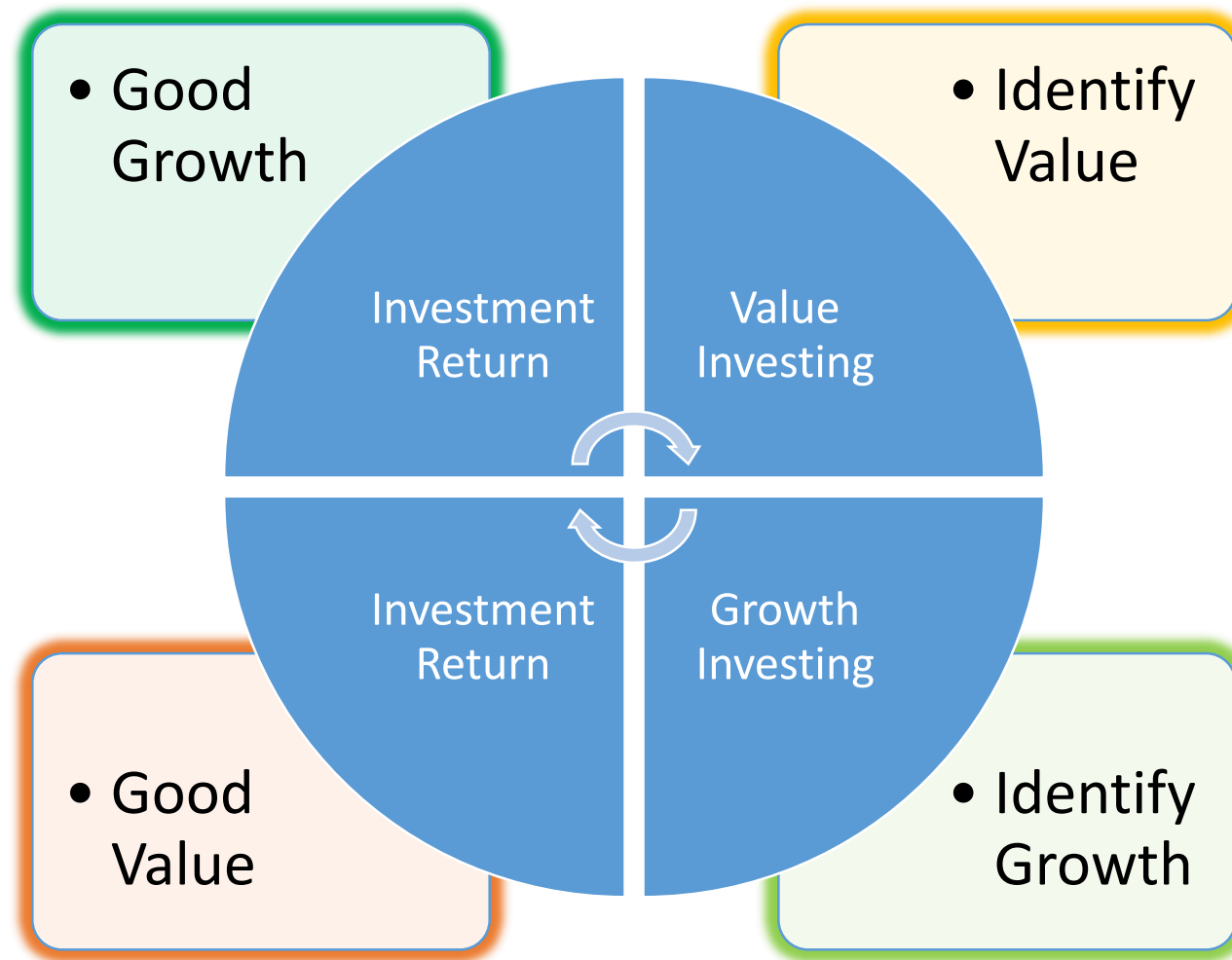
Growth Investing

- Identify shares whose future returns are expected to grow at a fast rate
- Short run and long run high growth rates from sales and EPS
- High profit margin
- Increase in projected earnings for both three and five years
- Financial soundness of the company
- Quantum of the company's debt obligations
- Quality of management

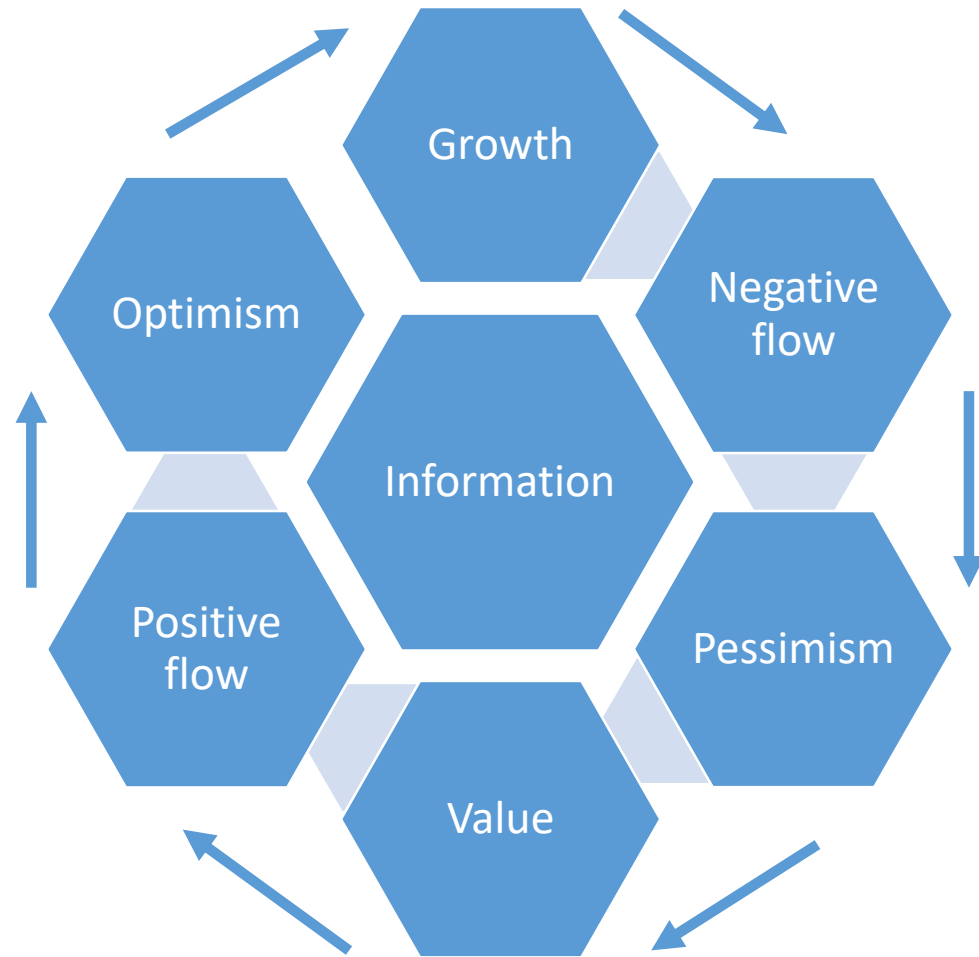
Value Investing

- Selection of shares at attractive low prices
- Share prices have been unfairly beaten down in the market
- Market prices that are below the estimated current economic value of tangible and intangible assets
- P/E Ratios
- Ratio of price to net working capital
- P/B Ratios
- Price to Cash Flow Ratios
- Dividends
- Private Market Value
- Managerial Holding

Investment Style



Flow of Investment Style



Problems in Style Investing

- Growth orientation
 - Customer Loyalty Risk
 - Merger Risk
 - Regulation Risk
 - Market Price Risk
- Value orientation
 - Governance Risk
 - Shareholding Structure Risk
 - Performance Risk
 - Market Price Risk

Efficient Market Theory

- If markets are efficient
 - It is not possible for an investor to outperform the market on a sustained basis
- Levels of information efficiency
 - Strong form of market efficiency
 - Semi-strong form of market efficiency
 - Weak form of market efficiency

Random Walk Theory

- In an efficient market
 - Prices are expected to show random movements
- Possibility of portfolio diversification subdivides the total risk into security (portfolio) specific risk and market risk.

$$\sigma_p^2 = \left(\beta_{p,i}^2 \times \sigma_i^2 \right) + \left(\sigma_{\varepsilon,p}^2 \right)$$

Market
Risk

+

Security
Specific Risk

- Total risk = Un-diversifiable risk + Diversifiable risk

Portfolio variance;	beta of the portfolio with the market;	market variance;	portfolio error variance
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σ_p^2

$\beta_{p,i}^2$

σ_i^2

$\sigma_{\varepsilon,p}^2$

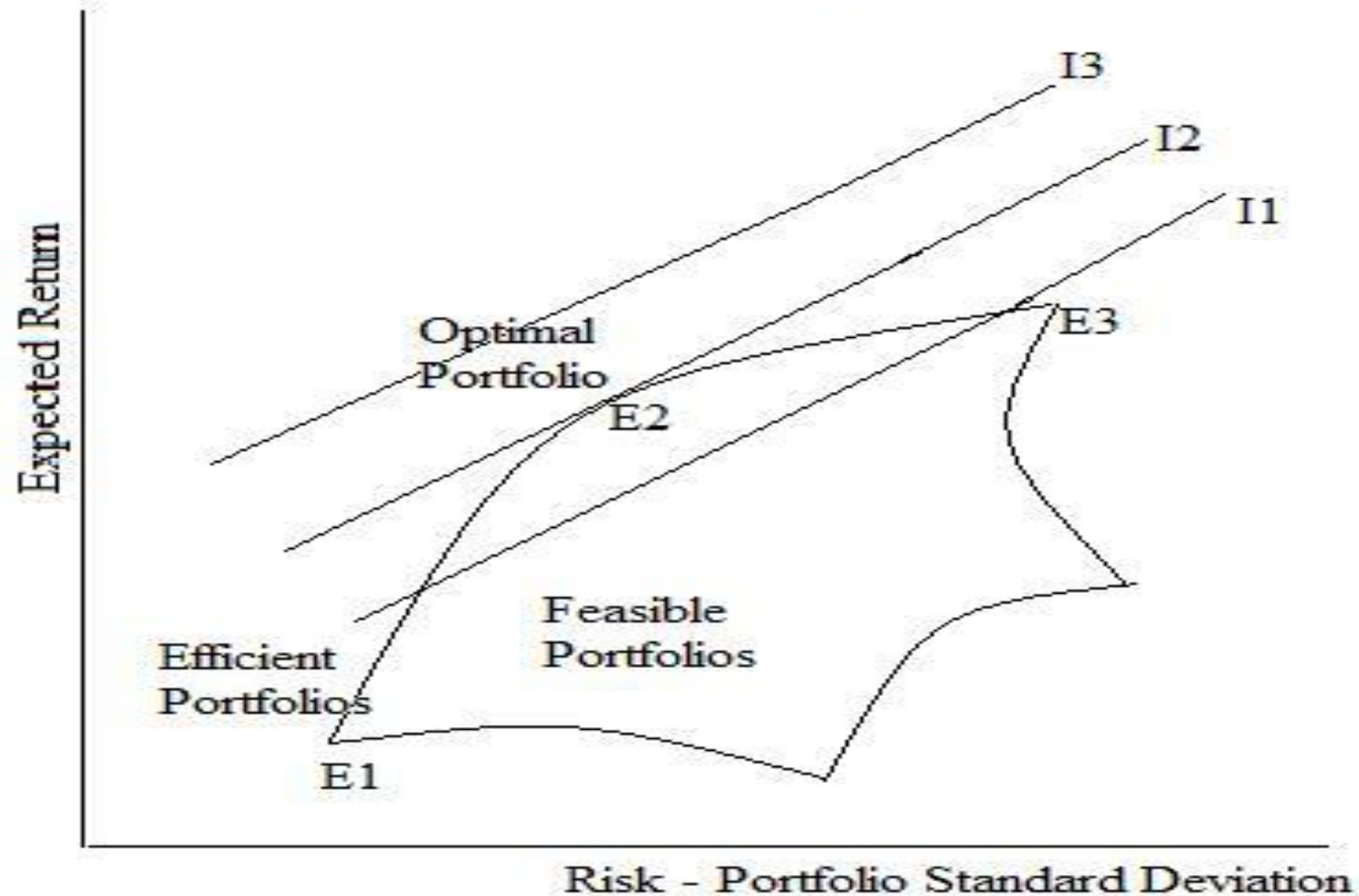
Markowitz Model

Assumptions

- Investors are risk averse
- Investors are intent on maximizing portfolio utility than its return
- All investors have the same time period as the investment horizon
- Portfolio with the investor's desired risk-return combination is feasible in the market
- Investment behaviour is to be distinguished from that of speculative behaviour of investors

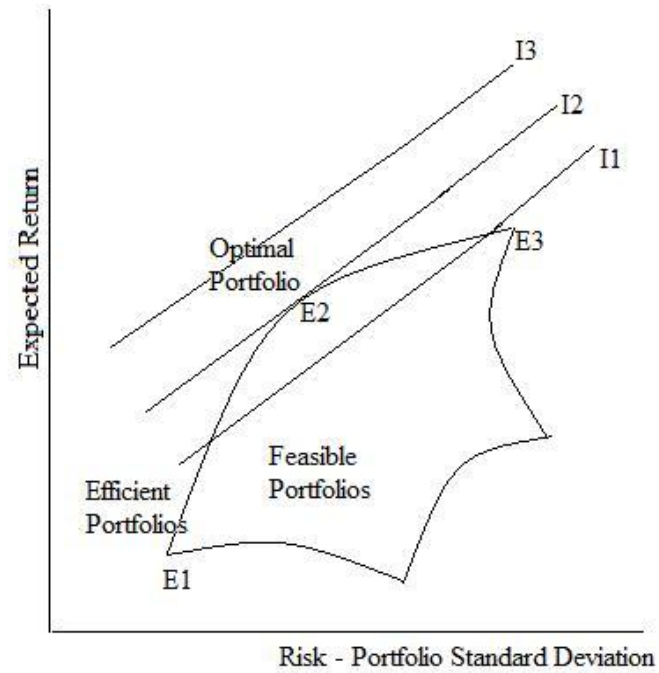
Efficient Frontier

Markowitz Model: Optimal Portfolio

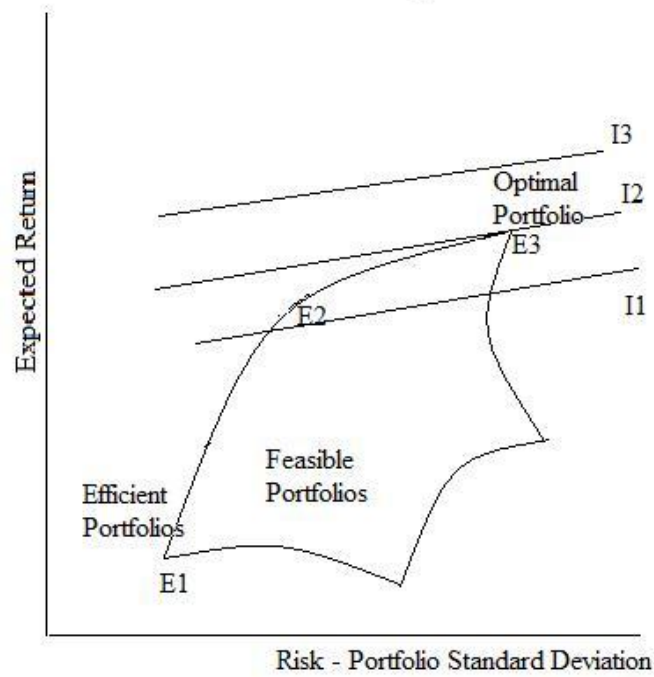


Portfolio Selection

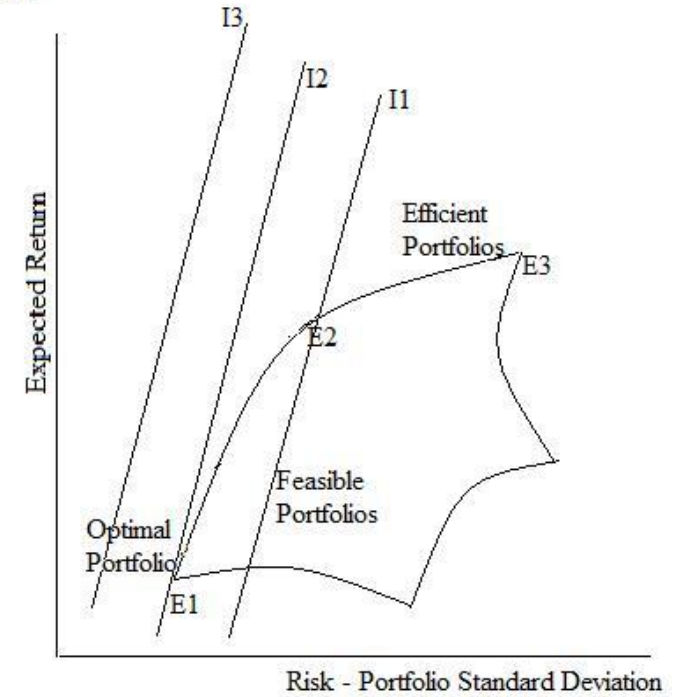
Markowitz Model: Optimal Portfolio



(a)



(b)

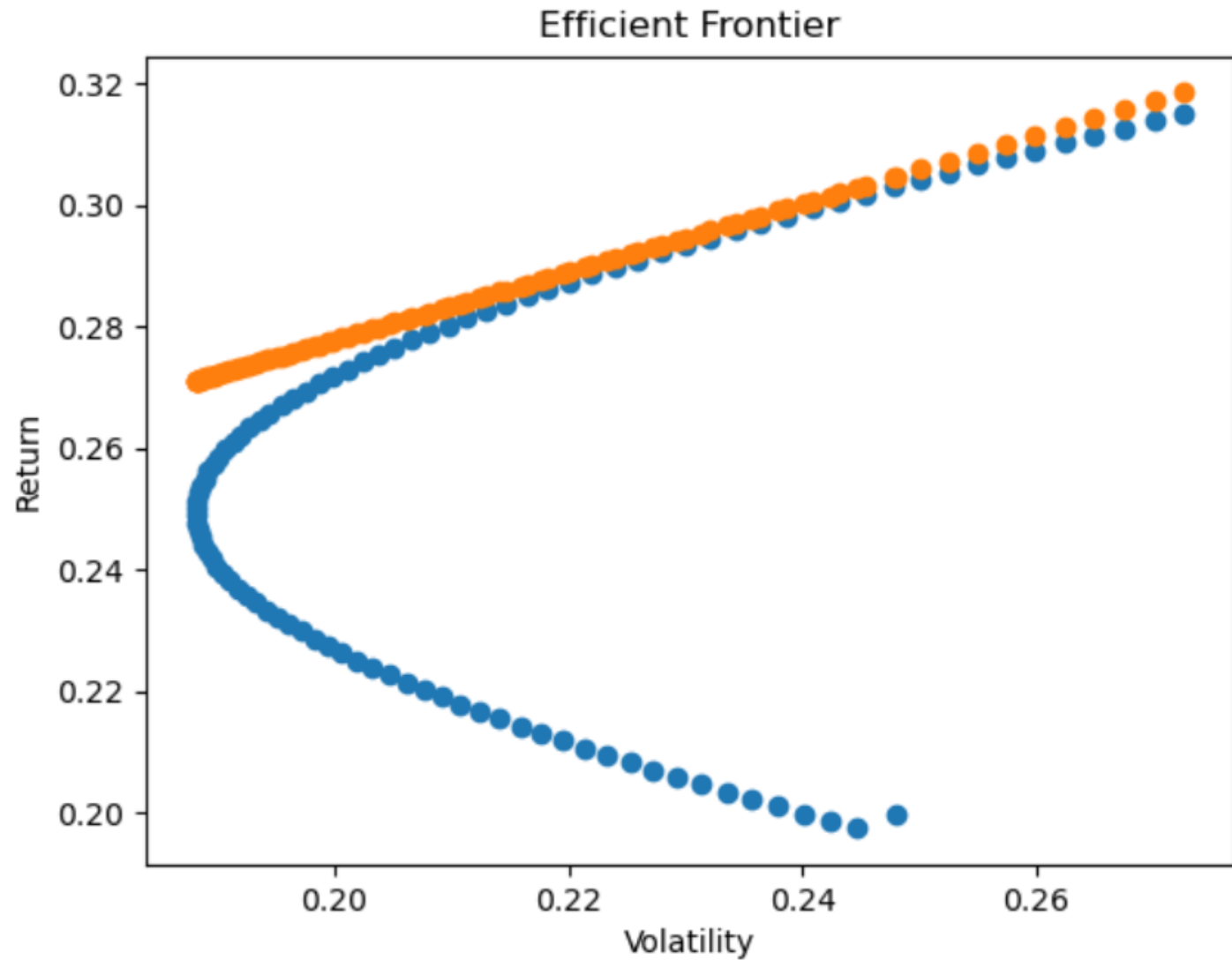


(c)

Three Security Portfolio

Security		A	B	C
	Return	Covariance Matrix		
A	0.15	0.0324	0.00216	-0.0081
B	0.17	0.00216	0.04	0.0125
C	0.2	-0.0081	0.0125	0.0625

Optimal Portfolio



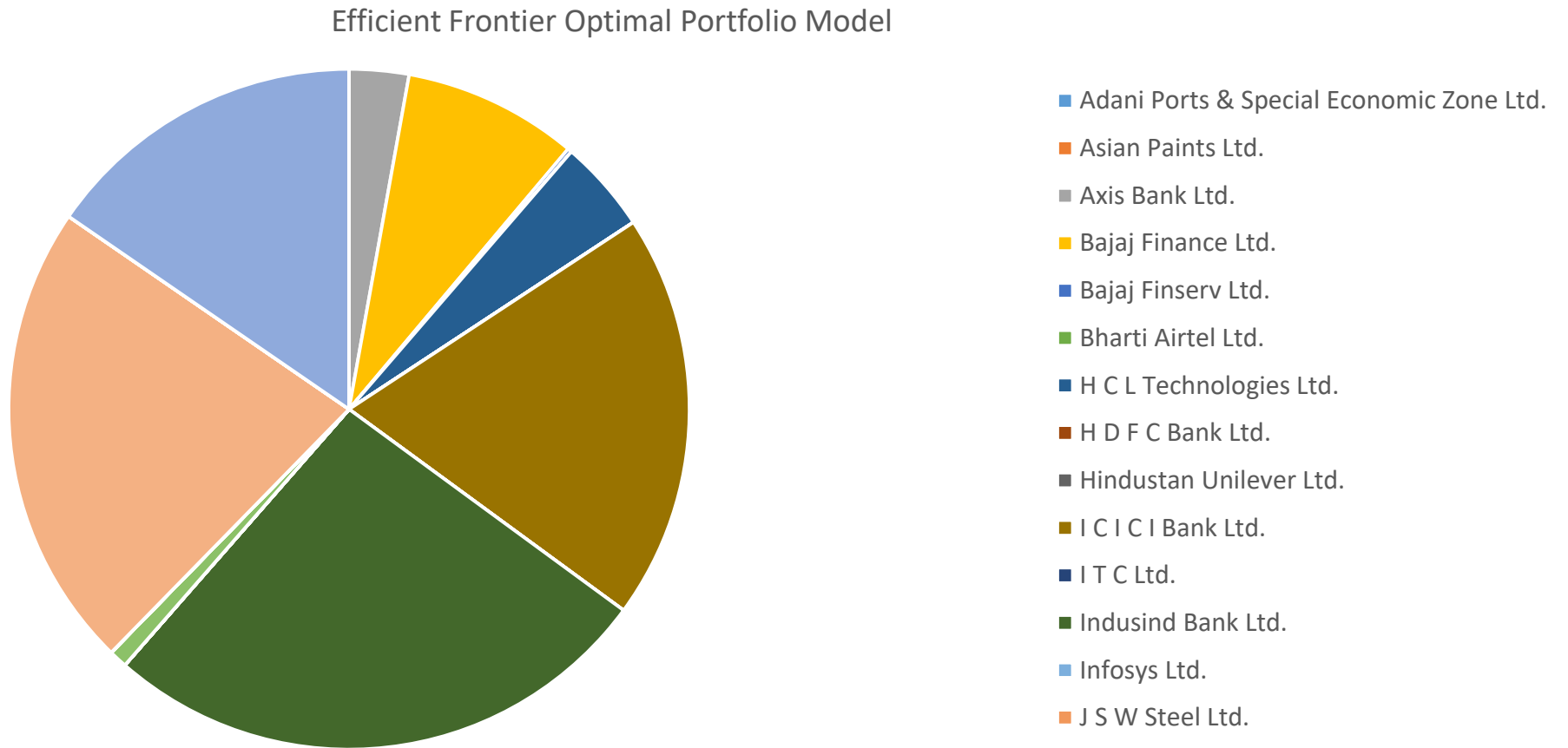
Minimum Risk Portfolio

	Weight A	Weight B	Weight C	Portfolio Return	Portfolio Risk	CML
	0.36	0.35	0.29	0.1715	0.1255	0.1698
	0.37	0.36	0.27	0.1707	0.1248	0.1696
	0.38	0.37	0.25	0.1699	0.1243	0.1694
Min Risk Portfolio	0.39	0.38	0.23	0.1691	0.1240	0.1694
	0.4	0.39	0.21	0.1683	0.1241	0.1694
	0.41	0.4	0.19	0.1675	0.1243	0.1694
	0.42	0.41	0.17	0.1667	0.1248	0.1696

Optimal Portfolio

	Weight A	Weight B	Weight C	Portfolio Return	Portfolio Risk	CML
	0.06	0.05	0.89	0.1955	0.2236	0.1966
	0.07	0.06	0.87	0.1947	0.2190	0.1953
Optimal Portfolio	0.08	0.07	0.85	0.1939	0.2144	0.1940
	0.09	0.08	0.83	0.1931	0.2099	0.1928
	0.1	0.09	0.81	0.1923	0.2054	0.1916
	0.11	0.1	0.79	0.1915	0.2010	0.1904
	0.12	0.11	0.77	0.1907	0.1967	0.1892

Optimal Portfolio Model (Sensex constituents)



Markowitz Portfolio

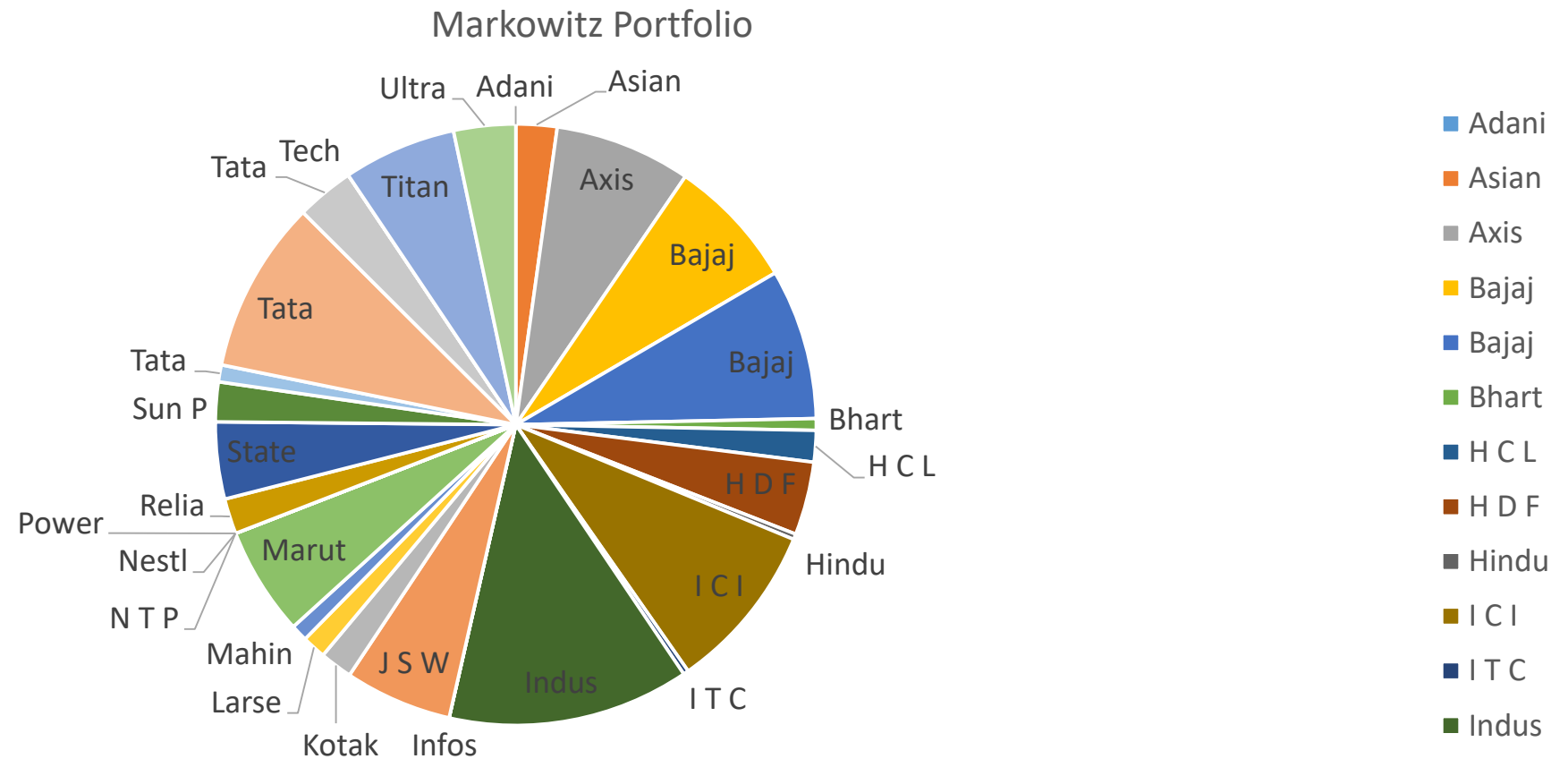
- Risk adjusted return (utility)
= Portfolio expected return - Risk penalty

$$\text{Risk Penalty} = \frac{\text{Risk squared}}{\text{Risk tolerance}}$$

<i>Portfolio</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
Return (%)	10	18	25	28	30	27	27	30	35	13
Risk (%)	15	20	25	24	29	25	23	28	30	17
Risk penalty	7.5	13.33	20.83	19.2	28.03	20.83	17.63	26.13	30	9.63
Utility	2.5	4.67	4.17	8.8	1.97	6.17	9.37	3.87	5	3.37
Rank	9	5	6	2	10	3	1	7	4	8

Risk tolerance: 30%

Markowitz Utility Model (Sensex constituents)



Factor Models

- Identify significant variable influencing the portfolio value variations
- Market model depicts the most important factor influencing the portfolio
- Economic factors are gross domestic product, production growth rate, etc.
- Industry specific factors are price index, industry demand, etc.

Sharpe's Single Index Model

- Desirability of a security for its inclusion is directly related to its excess return to beta ratio

$$\frac{R_i - R_f}{\beta_i}$$

- Where

R_i = expected return on security i

R_f = return on a riskless security

β_i = beta of security i

- Ranking order gives the best securities that are to be selected sequentially for the portfolio.

Sharpe's Single Index Model

- Weight of securities in the optimal portfolio would be based on the final cumulative return of all securities in relation to estimated risk.
- Computation is based on the cut-off rate

- where

σ_m^2 = market variance

R_i = security return

R_f = risk free return

β_i = security beta

σ_{ei}^2 = security error variance

- Final cumulative rate C^* is one where all the securities have been included in the portfolio.

$$C_i = \frac{\sigma_m^2 \sum_{i=1}^i \frac{(R_i - R_f) * \beta_i}{\sigma_{ei}^2}}{1 + \sigma_m^2 \sum_{i=1}^i \frac{\beta_i^2}{\sigma_{ei}^2}}$$

Sharpe's Single Index Model

- Percentage of investment in each of the securities in a portfolio with optimal C^* cut-off rate

$$\omega_i = \frac{Z_i}{\sum_{i=1}^n Z_i}$$

- Optimal weight of each security in the selected portfolio is expressed as the excess return adjusted to the estimated cut-off return in relation to the standard error of the security.

$$Z_i = \frac{\beta_i}{\sigma_{ei}^2} \left(\frac{R_i - R_f}{\beta_i} - C^* \right)$$

R_i = security return
 R_f = risk free return
 β_i = security beta
 σ_{ei}^2 = security error variance
 C^* = Cut-off value

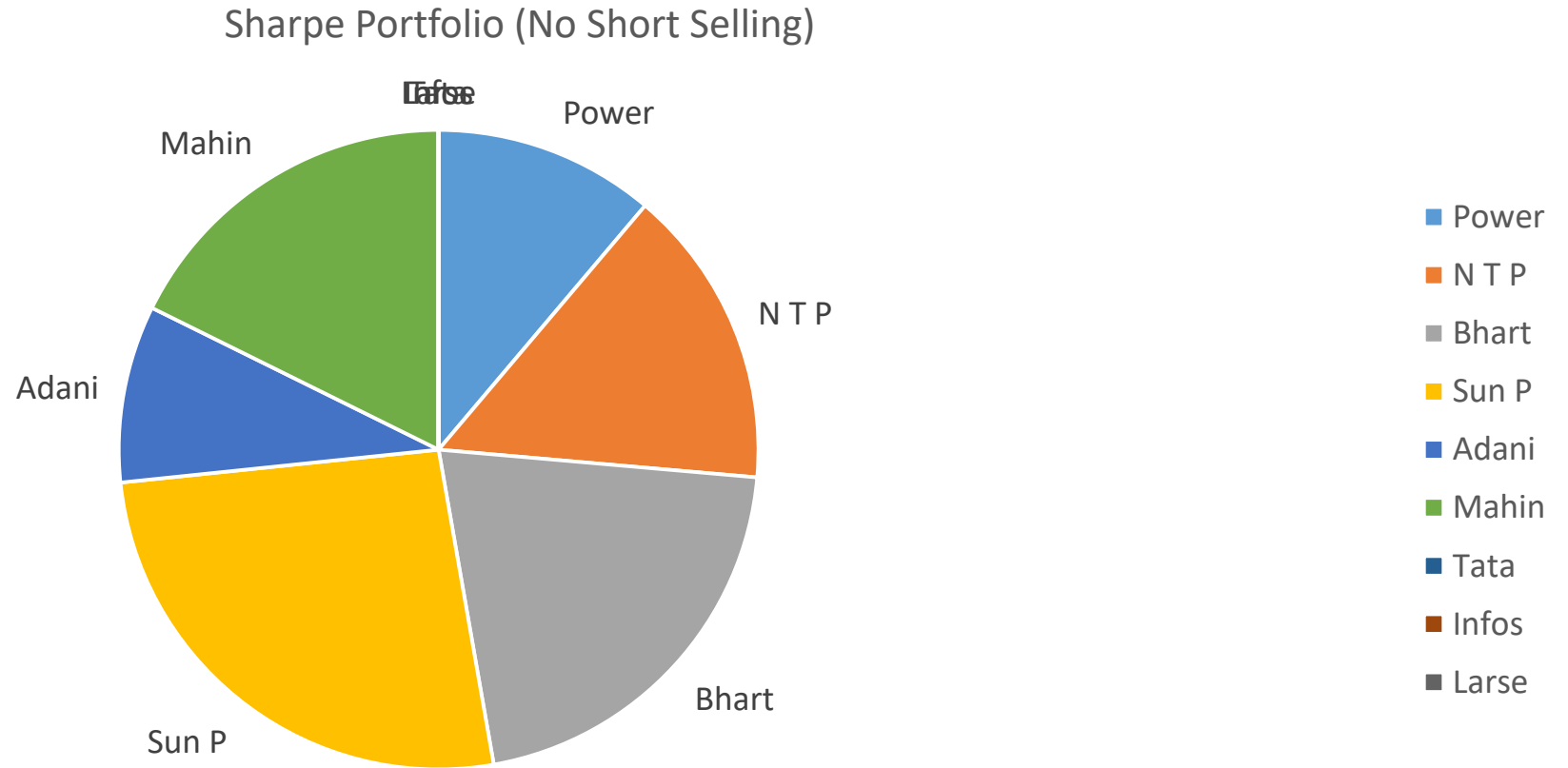
Securities Arranged in Descending Order

Security	Return	Beta	Error	Security Variance	Error variance	Impact of Systematic risk on return	Impact of Security risk on return	Cumulative return to estimation risk	Expected return to estimation risk
F	0.16	0.6	0.14	0.0421	0.0196	0.1833	5.6122	3.3673	0.2105
J	0.2	1.5	0.16	0.1662	0.0256	0.1000	5.8594	12.1564	0.7598
D	0.18	1.3	0.2	0.1456	0.04	0.1000	3.2500	16.3814	1.0238
G	0.13	0.8	0.16	0.0656	0.0256	0.1000	3.1250	18.8814	1.1801
E	0.14	1.02	0.15	0.0875	0.0225	0.0882	4.0000	22.9614	1.4351
C	0.13	1.2	0.17	0.1189	0.0289	0.0667	2.7682	26.2832	1.6427
H	0.14	1.5	0.13	0.1575	0.0169	0.0600	5.3254	34.2714	2.1420
B	0.15	1.8	0.16	0.2281	0.0256	0.0556	3.9063	41.3026	2.5814
I	0.11	1.2	0.14	0.1096	0.0196	0.0500	3.0612	44.9761	2.8110
A	0.12	1.5	0.15	0.1631	0.0225	0.0467	3.1111	49.6428	3.1027

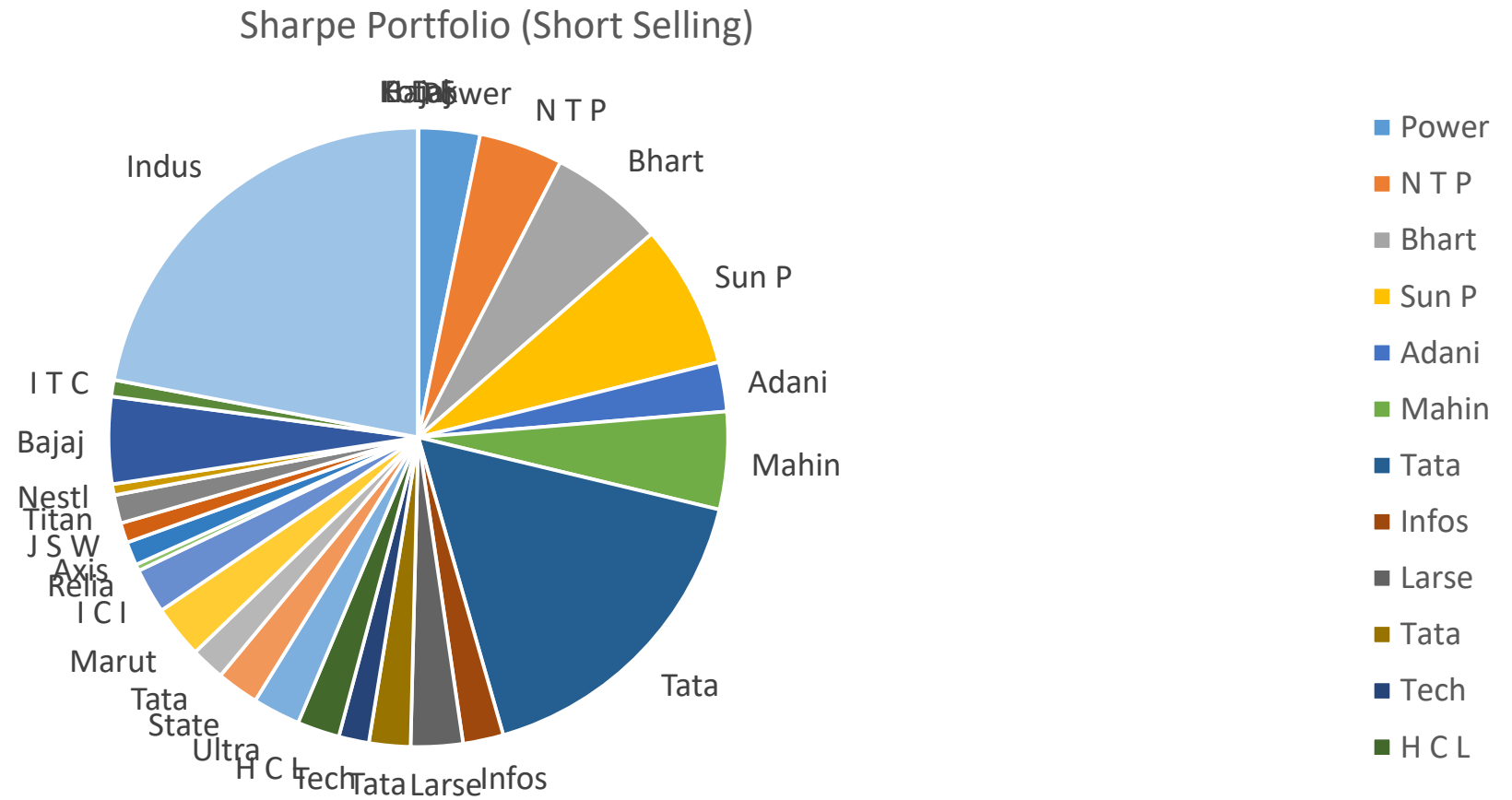
..... Securities Arranged in Descending Order

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Sharpe's Model (No Short Selling) (Sensex constituents)



Sharpe's Model (Short Selling) (Sensex constituents)



Arbitrage Pricing Theory

Assumptions

- Investors prefer more returns to fewer returns
- Investors are risk averse
- Investors have homogeneous risk expectations
- Capital markets do not have any transaction cost and there are no taxes

Arbitrage Pricing Theory

- Arbitrage pricing theory for a single factor model

$$R_i = E_i + b_{i1} \delta_1$$

- where

R_i = expected return for asset i

E_i = is the return for a zero beta portfolio

B_{i1} = is the factors' risk premium

δ_1 = the sensitivity of the security to the specific risk factor

Arbitrage Pricing Theory

- Arbitrage pricing theory for more than one factor model

$$R_i = E_i + b_{i1}\delta_1 + b_{i2}\delta_2 + \dots b_{ik}\delta_k + e_i$$

- where

for $i = 1$ to N

R_i = expected return for asset i

E_i = is the return for a zero beta portfolio

b_{i1} to b_{ik} = is the factors' risk premium

δ_1 to δ_k = The sensitivity of the i^{th} asset to the specific risk factor

N = number of assets

k = number of factors

Arbitrage Relation

- Arbitrage Relation

$$V_{it} = V_{jt}; \text{ if } D_{i0} = D_{j0}, r_i = r_j; \text{ and } g_i = g_j$$

i : company that is being valued; j : comparable company; t : time

V_0 : Equity value at time 0; D_0 : Dividend at period 0;

g : Expected dividend growth rate; r : Required rate of return.

- Synthetic Arbitrage Relation

$$V_{it} = V_{jt} \left(F_{i\tau} \left(F_{j\tau} \right)^{-1} \right)$$

F : factor that is allowed to vary between companies proportionally to market value of equity

τ : time at which the factors are measured

Multi-factor Valuation

$$V_{it} = V_{jt} \sum_{n=1}^x \left(a_{nt} \left(F_{in\tau} \left(F_{jn\tau} \right)^{-1} \right) \right)$$

n : different factors; x : total number of factors

a_{nt} : weighting of factor n at time t ;

$$\sum_{n=1}^x a_{nt} = 1 \text{ for all } t.$$

Fama-French Three Factor Model

- Market return
- Size of firms: SMB (small minus big)
- SMB (Small Minus Big) is the average return on the three small portfolios minus the average return on the three big portfolios,
- Book-to-market ratio: HML (high minus low)
- HML (High Minus Low) is the average return on the two value portfolios minus the average return on the two growth portfolios,

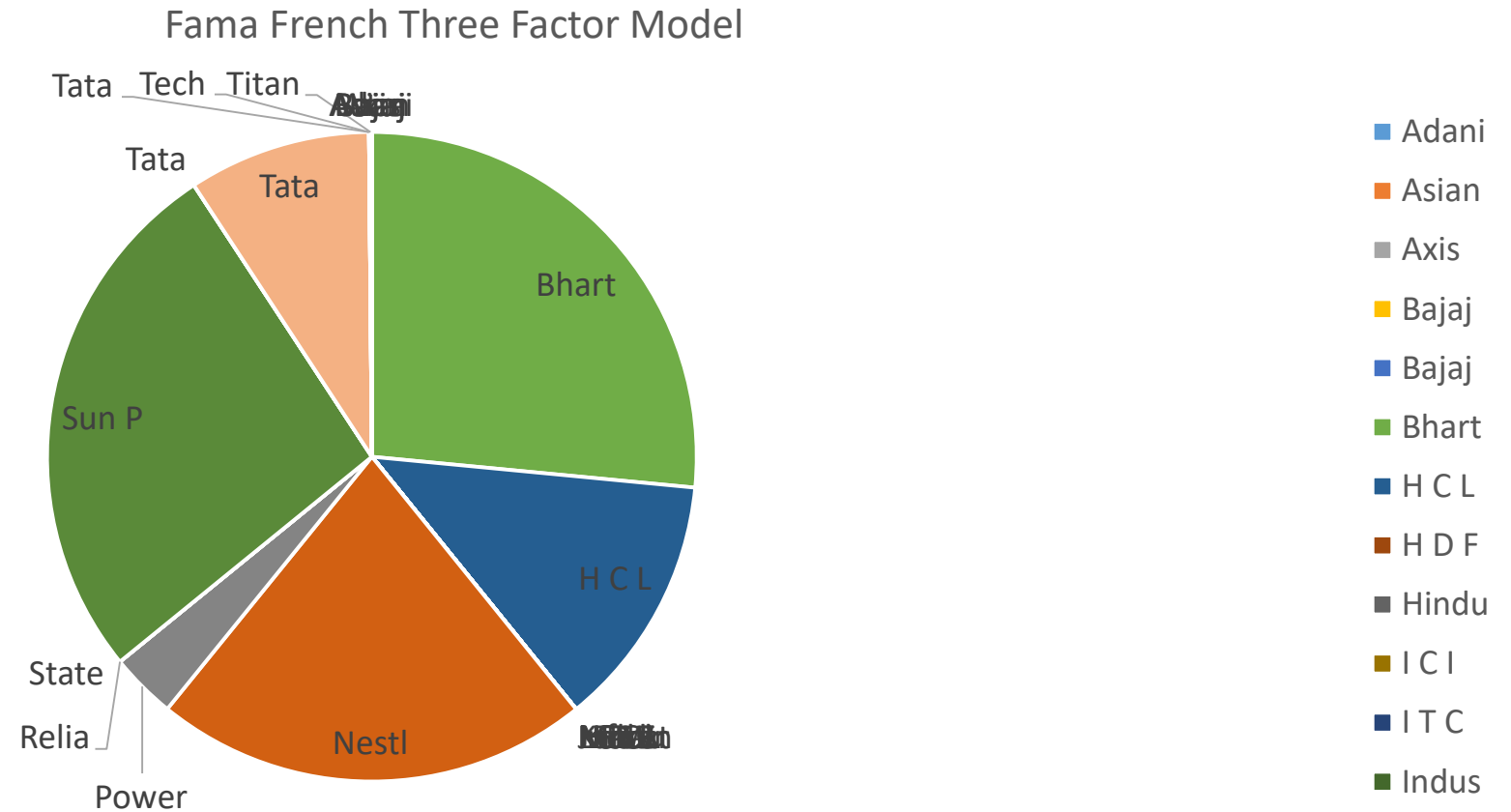
Three Factors

- **Market return** = $R_m - R_f$, excess return on the market
- **SMB = Small Minus Big**
- $\frac{1}{3} (\text{Small Value} + \text{Small Neutral} + \text{Small Growth}) - \frac{1}{3} (\text{Big Value} + \text{Big Neutral} + \text{Big Growth})$
- **HML = High Minus Low**
- $\frac{1}{2} (\text{Small Value} + \text{Big Value}) - \frac{1}{2} (\text{Small Growth} + \text{Big Growth})$

Three Factor Model

- $R_{it} - R_{ft} = \alpha_{it} + \beta_1(RM_t - R_{ft}) + \beta_2(SMB_t) + \beta_3(HML_t) + \varepsilon_t$
- R_{it} = Total return of a stock or portfolio i at time t
- R_{ft} = Risk free rate of return at time t
- RM_t = Total market portfolio return at time t
- $R_{it} - R_{ft}$ = Expected excess return
- $RM_t - R_{ft}$ = Excess return on the market portfolio (index)
- SMB_t = Size premium (small minus big)
- HML_t = Value premium (high minus low)
- $\beta_{1,2,3}$ = Factor coefficients

Fama French Model (Sensex Constituents)



Portfolio Model Comparison

